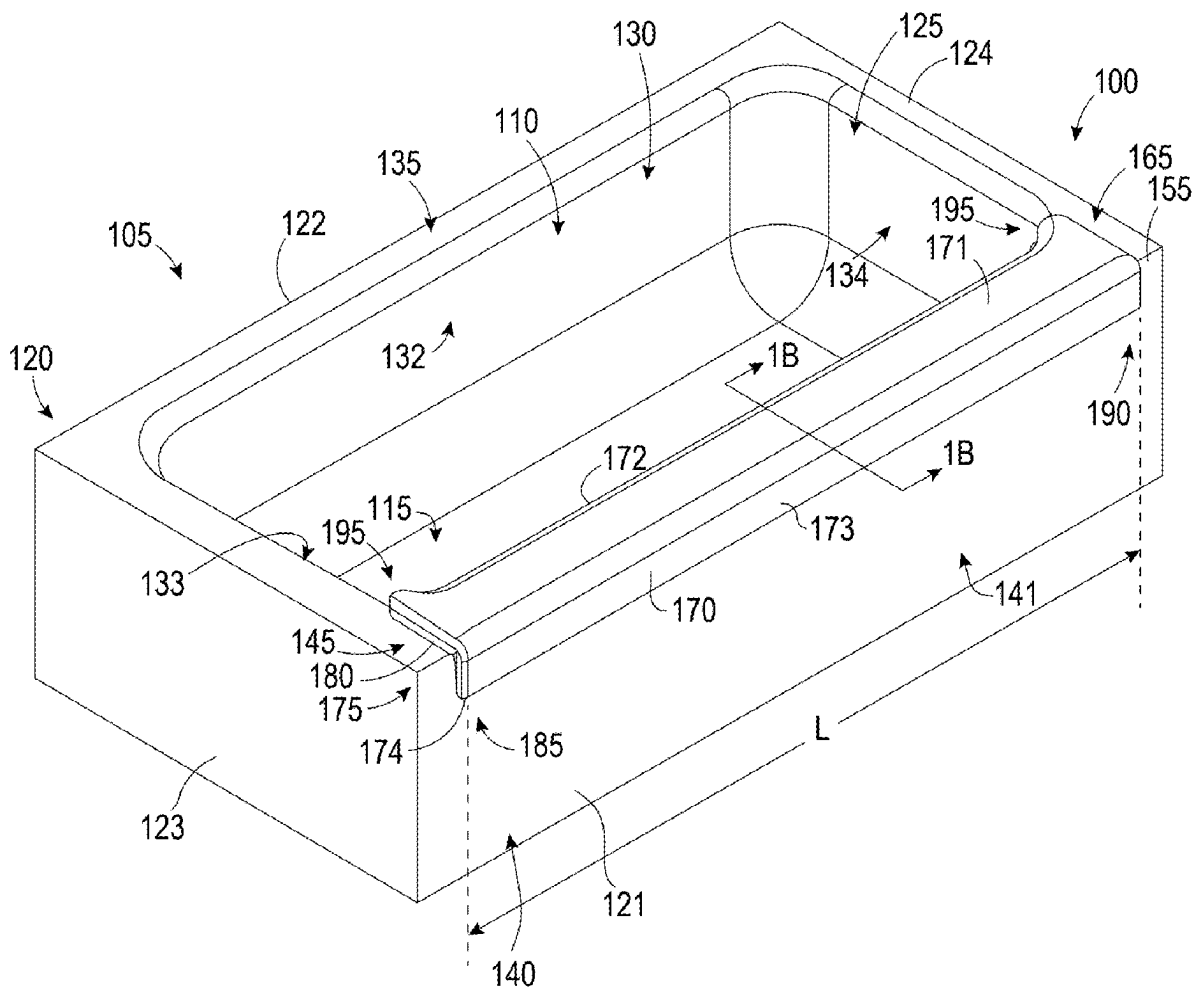


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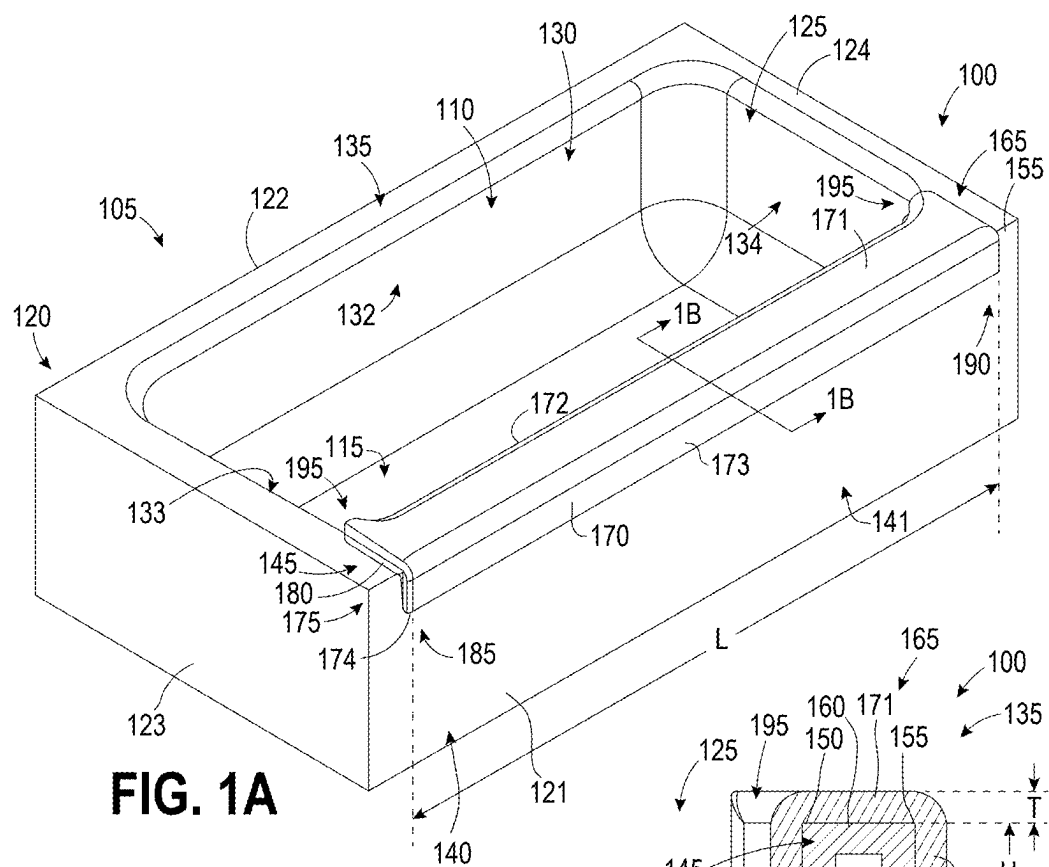


FIG. 1A

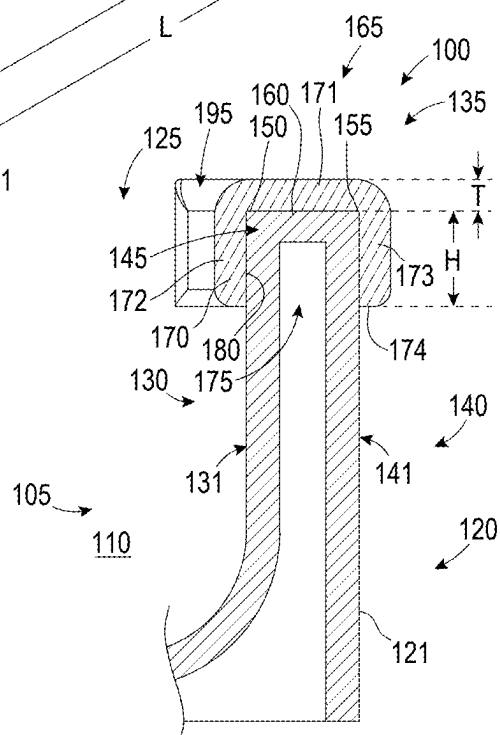
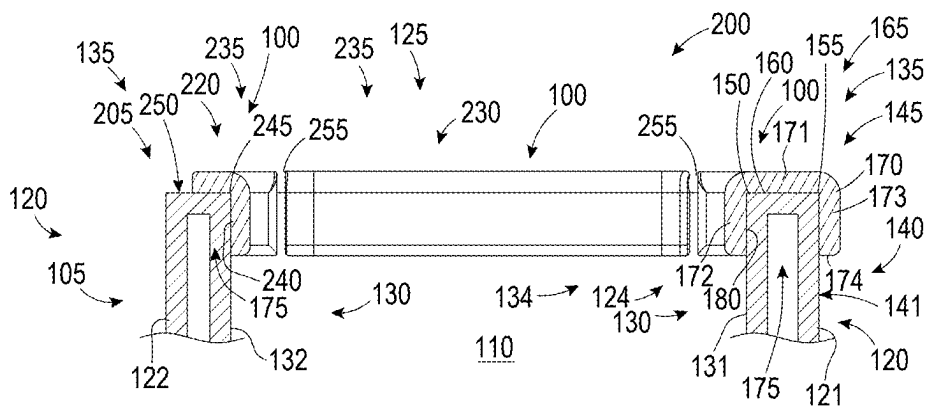
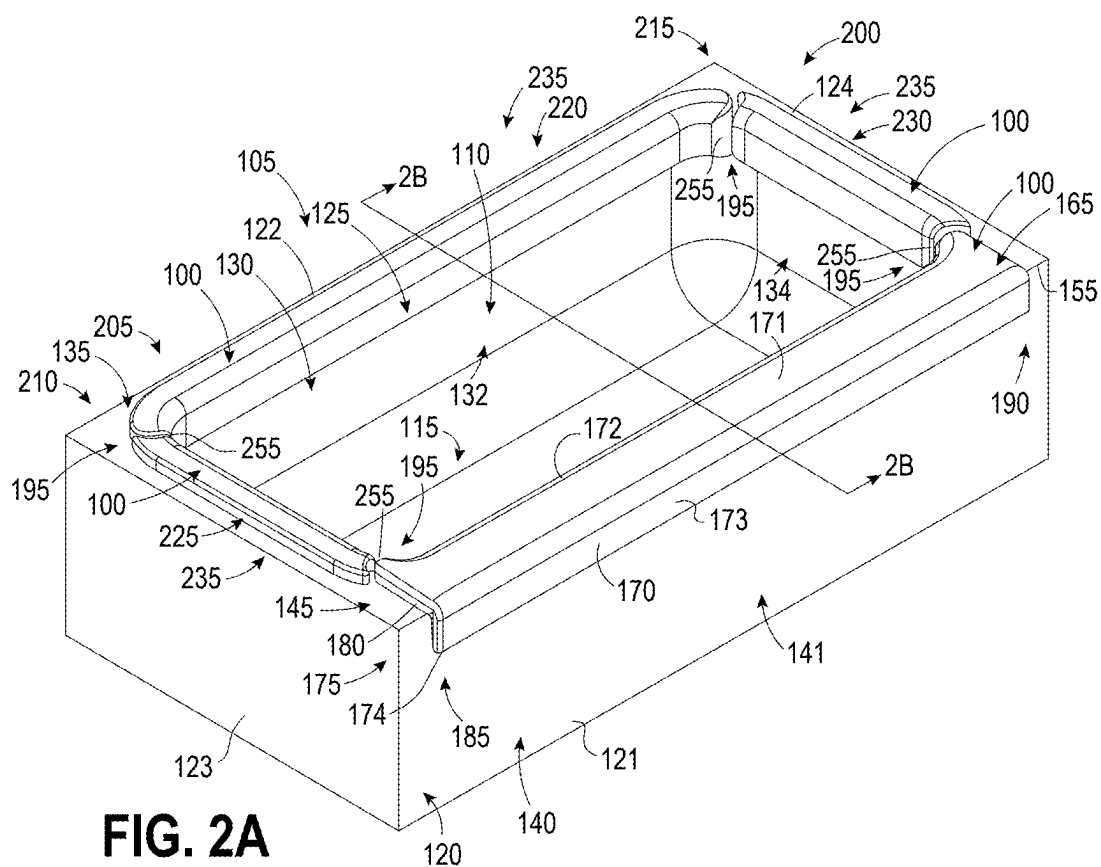


FIG. 1B



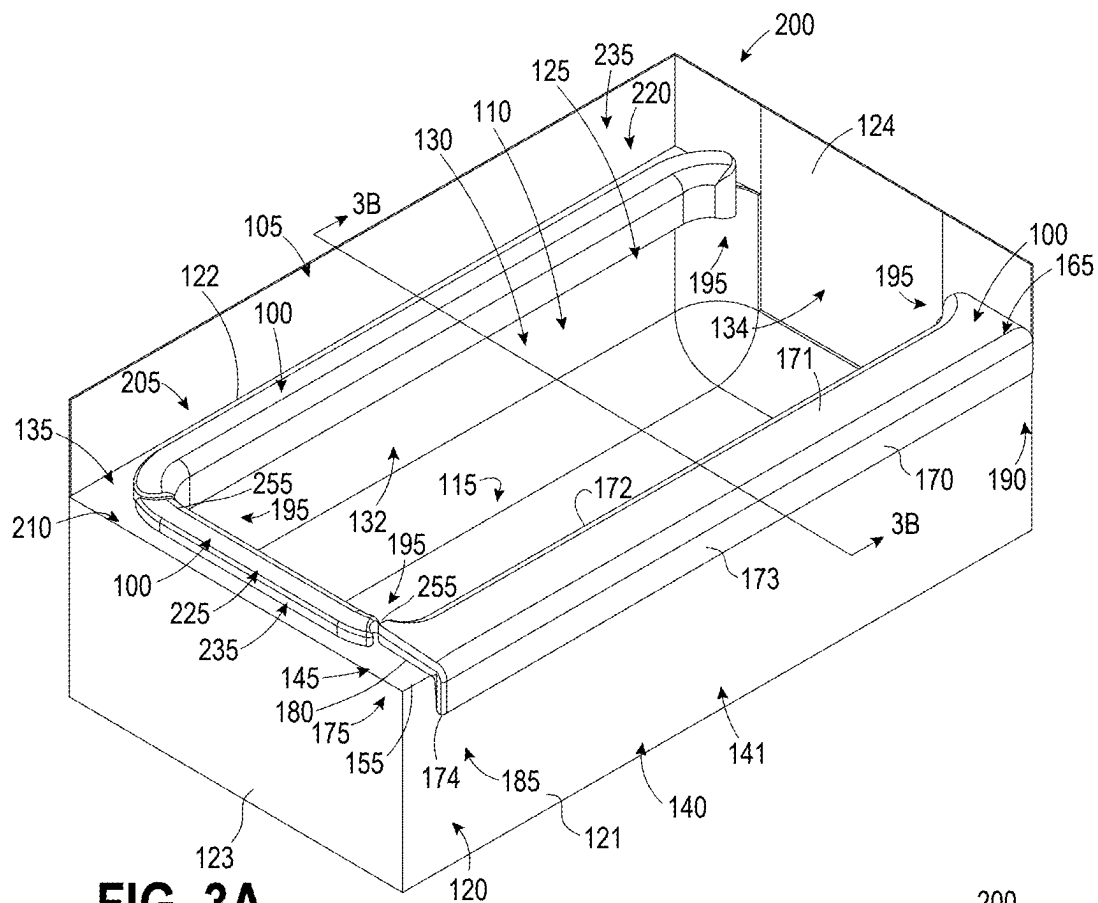


FIG. 3A

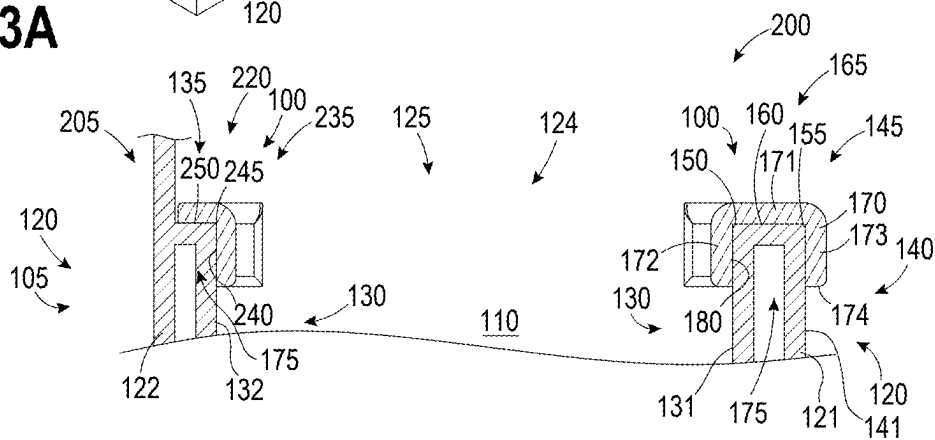


FIG. 3B

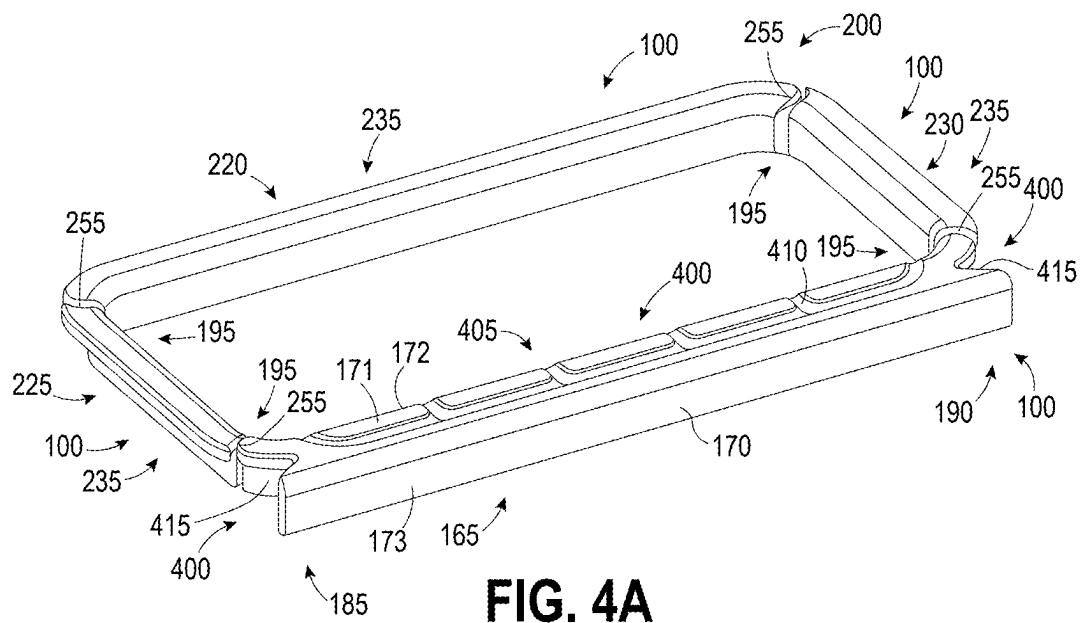


FIG. 4A

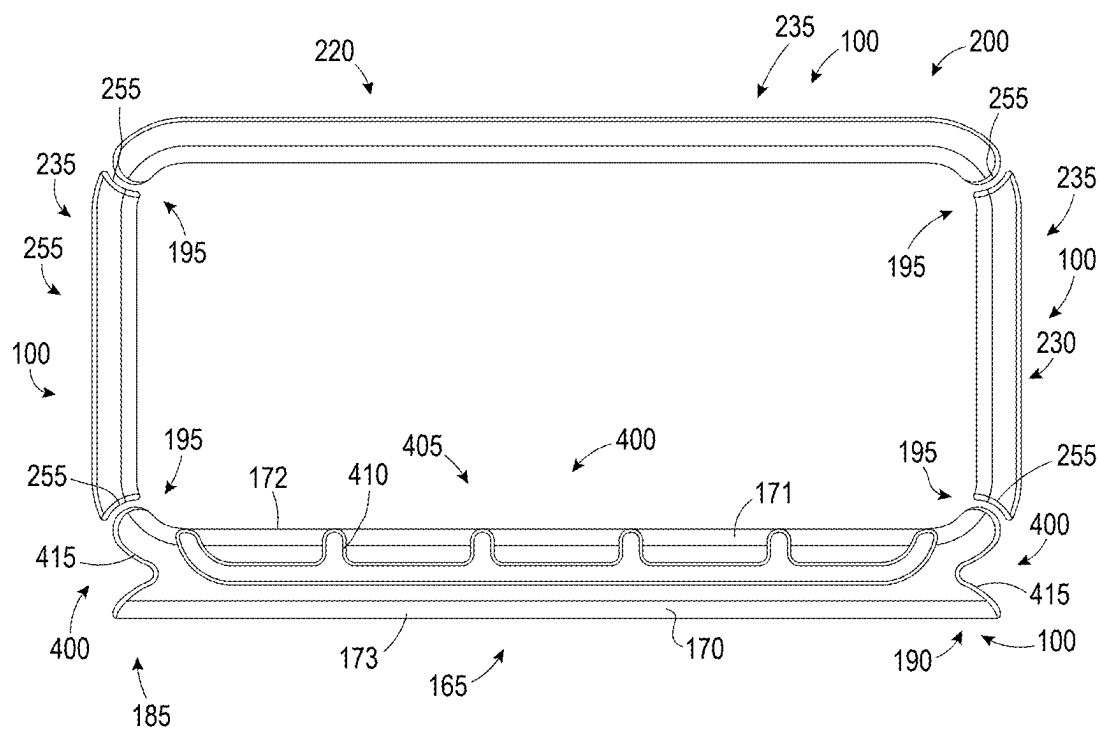
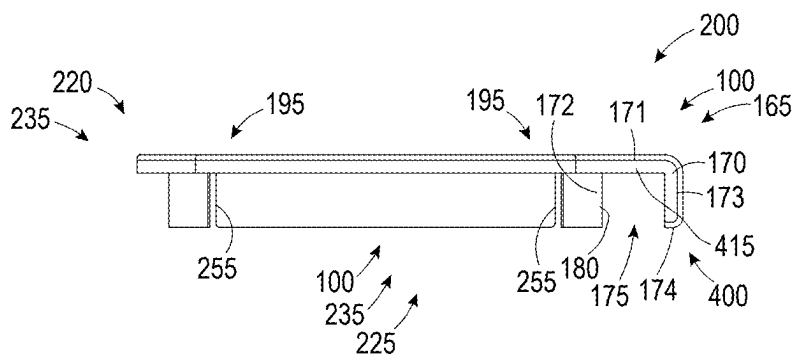
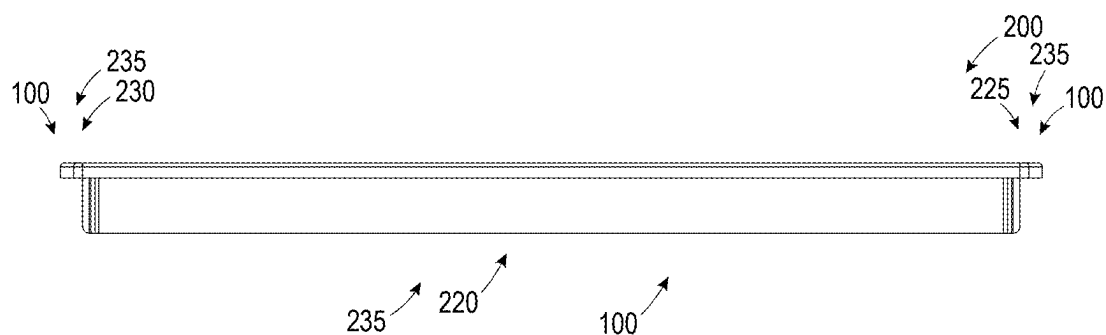
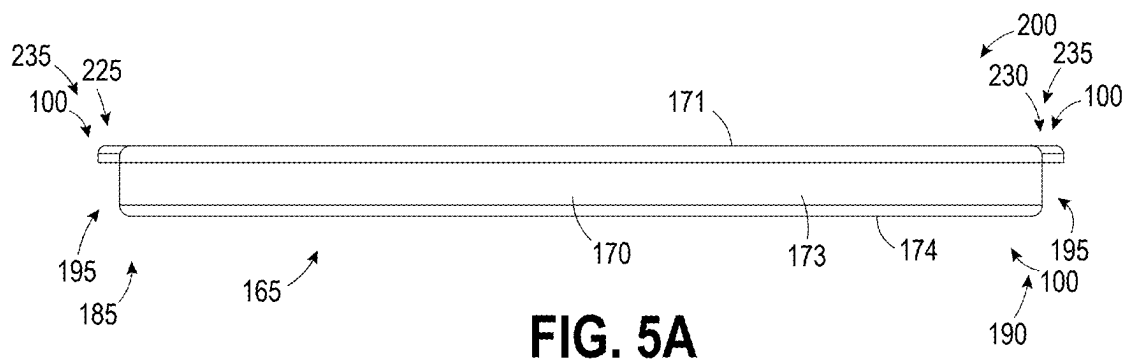


FIG. 4B



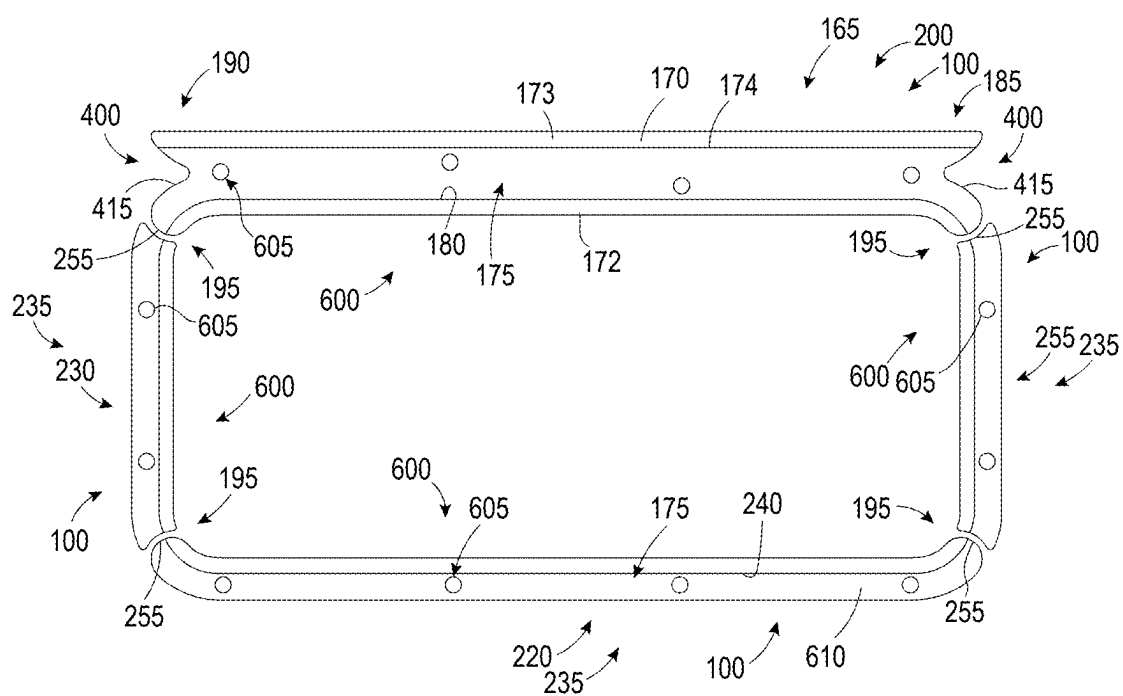


FIG. 6A

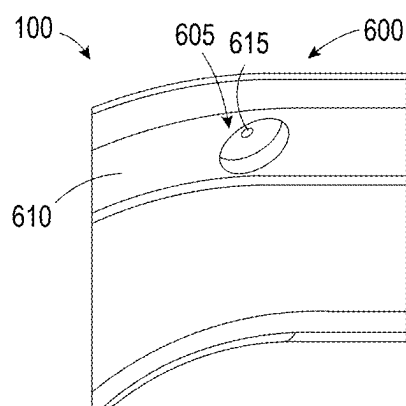


FIG. 6B

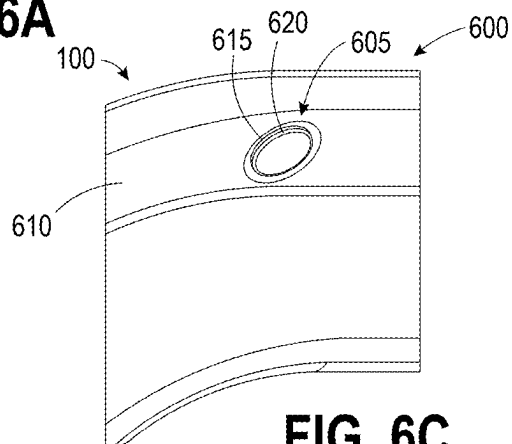


FIG. 6C

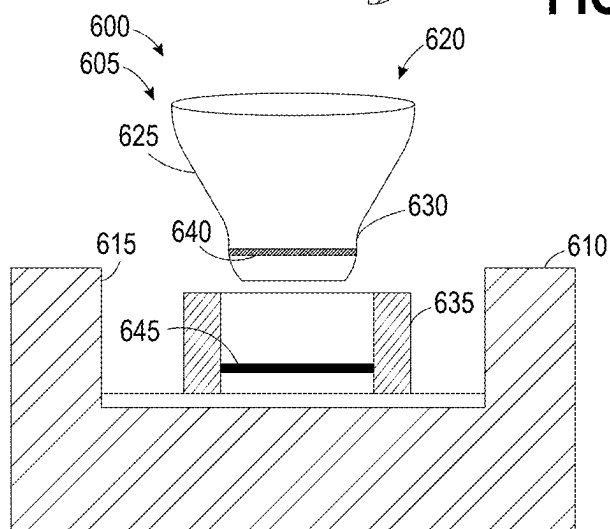


FIG. 6D

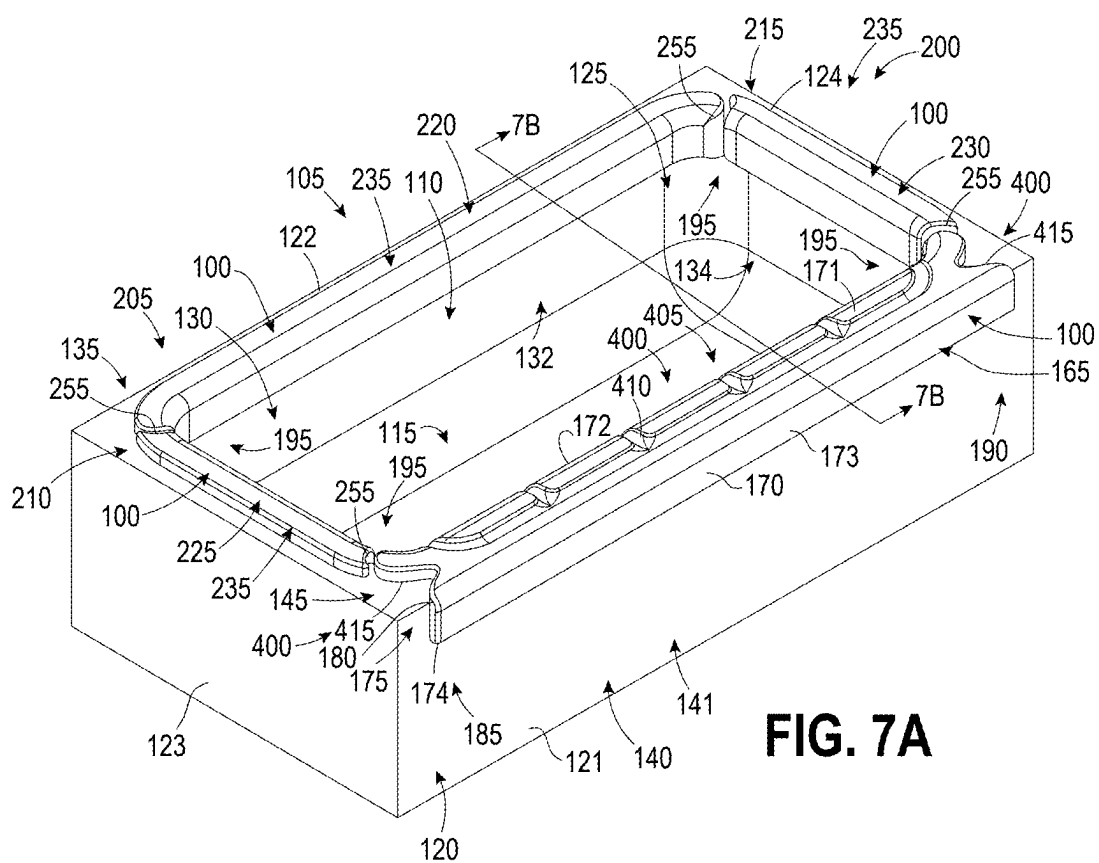


FIG. 7A

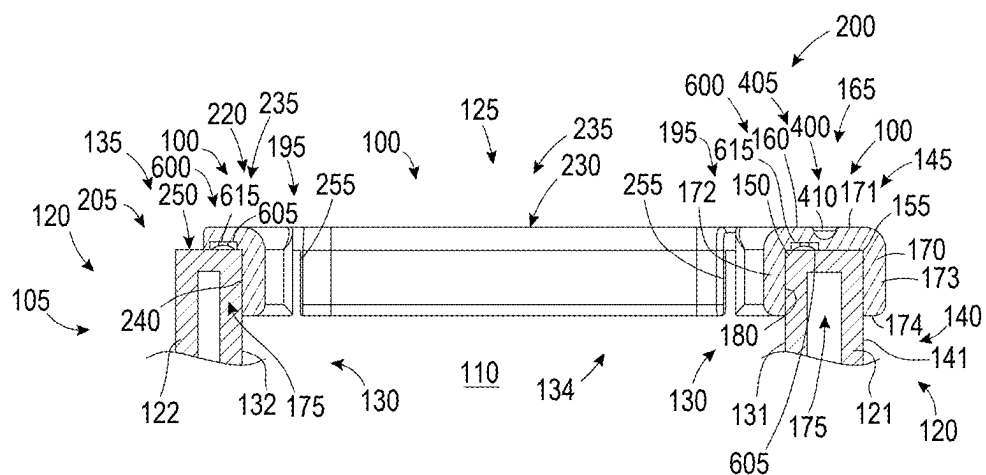


FIG. 7B

BATHTUB BUMPER

PRIORITY

[0001] The present application claims the benefit of domestic priority based on U.S. Provisional Patent Application 63/557,389 filed on Feb. 23, 2024, the entirety of which is incorporated herein by reference.

BACKGROUND

[0002] The bathroom, particularly in and around a bathtub, can be a dangerous place.

[0003] More than 200,000 injuries occur in bathrooms each year according to a study by the Center for Disease Control and Prevention. Of those, about two-thirds occur in and around the bathtub or shower. The slick and/or wet surfaces in a bathroom can make it easy for an individual to slip and fall, and exposed hard surfaces and edges create hazards for the individual on their way down. Often the injuries are serious. For example, more than 30% of the injuries resulting from slips and falls in the bathroom were to the head and neck region, more than 17% resulted in fractures, and more than 9% resulted in internal injuries. The danger is especially high for young children, the physically disabled, and the elderly who are more likely to lose their footing and less likely to be able to control their fall once it has begun. Very young children being bathed by a caregiver are also particularly vulnerable to bathtub injuries due to their erratic and unpredictable movements which can cause them to slip away from a caregiver's control, especially when wet.

[0004] One common approach to reducing the number of slip and fall injuries in a bathtub is the use of non-slip, traction increasing mats or strips. While useful in some instances, the use of non-slip surfaces has its limitations. Sometimes a user will forget to place a non-slip mat in a bathtub, and other times a user may not secure the mat properly when they do remember. Also, non-slip mats or strips often fail to cover an entire bathtub standing surface, leaving vulnerable, treacherous locations that can be slick when stepped on. In addition, a non-slip surface in the bathtub does not keep someone from slipping on the bathroom floor or from slipping while getting into or out of the bathtub. Furthermore, an individual can fall for reasons other than slipping, such as by losing their balance or leaning against an unsecure surface. Non-slip surfaces also do not help in cases when a caregiver drops or loses control of a young or elderly person in their care.

[0005] Heretofore there have not been adequate solutions for making a bathroom surface, such as a surface in or on a bathtub, less dangerous in event of a slip or fall. Sometimes an individual will cover hard surfaces or corners with an object such as a towel, but a towel does not offer much protection and can easily fall off or be pulled off. Elbow pads are available that rest on the front rim of a bathtub to cushion a caregiver's elbows while they bend over the front rim to bathe a child. However, existing elbow pads only cover a small portion of the potentially dangerous surfaces of the bathtub. There are also large, plastic bathtub splash protectors or guards that can be positioned on bathtubs, but these devices offer no protection against falls. In addition, these devices provide additional hazards by exposing sharp edges and/or creating obstacles that have to be awkwardly navigated around, such as when getting in or out of the bathtub.

[0006] There is therefore a need for an improved apparatus for decreasing the risk and/or severity of injuries in and around a bathtub. There is further a need for a bathtub bumper that is adapted to cover one or more potentially dangerous surfaces of a bathtub in an improved manner, such as by providing greater coverage, by being easily and securably installable, by offering placement and positioning options and adaptability, and/or providing additional bathing-related features. There is further a need for a bathtub bumper system that is adapted to cover a plurality of potentially dangerous surfaces of a bathtub. There is a further need for a bathtub bumper system that offers flexibility, adaptability, and/or modularity.

SUMMARY

[0007] The present invention satisfies these needs. In one aspect of the invention, a bathtub bumper is provided that provides improved protection in and around a bathtub.

[0008] In another aspect of the invention, a bathtub bumper is provided that decreases the risk and/or severity of injuries in and around a bathtub.

[0009] In another aspect of the invention, a bathtub bumper is provided that is adapted to cover one or more potentially dangerous surfaces of a bathtub in an improved manner.

[0010] In another aspect of the invention, a bathtub bumper is provided that is adapted to cover one or more potentially dangerous surfaces of a bathtub in an improved manner, such as by providing greater coverage of the dangerous surface.

[0011] In another aspect of the invention, a bathtub bumper is provided that is adapted to cover one or more potentially dangerous surfaces of a bathtub in an improved manner, such as by being easily and securably installable onto a bathtub surface.

[0012] In another aspect of the invention, a bathtub bumper is provided that is adapted to cover one or more potentially dangerous surfaces of a bathtub in an improved manner, such as by offering placement and positioning options and adaptability for a particular bathtub design.

[0013] In another aspect of the invention, a bathtub bumper is provided that is adapted to cover one or more potentially dangerous surfaces of a bathtub in an improved manner, such as by reducing the likelihood of mold growth under the bathtub bumper.

[0014] In another aspect of the invention, a bathtub bumper is provided that is adapted to cover one or more potentially dangerous surfaces of a bathtub in an improved manner, such as by providing additional bathing-related features, such as a drainage channel and/or item storage.

[0015] In another aspect of the invention, a bathtub bumper system comprises a plurality of bathtub bumpers.

[0016] In another aspect of the invention, a bathtub bumper system comprises a first bathtub bumper and a second bathtub bumper different in shape or configuration than the first bathtub bumper.

[0017] In another aspect of the invention, a bathtub bumper comprises a front wall bathtub bumper that is sized and shaped to be positionable on a front wall rim portion of a bathtub.

[0018] In another aspect of the invention, a bathtub bumper comprises a front wall bathtub bumper that is sized and shaped to be positionable on a front wall rim portion of a

bathtub, wherein the front wall bathtub bumper covers both the front wall inner corner and the front wall outer corner.

[0019] In another aspect of the invention, a bathtub bumper comprises a front wall bathtub bumper that is sized and shaped to be positionable on a front wall rim portion of a bathtub, wherein the front wall bathtub bumper covers the front wall outer corner and at least a portion of the front wall exterior surface.

[0020] In another aspect of the invention, a bathtub bumper comprises a front wall bathtub bumper that is sized and shaped to be positionable on a front wall rim portion of a bathtub, wherein the front wall bathtub bumper covers the front wall outer corner and at least a portion of the front wall exterior surface and wherein the front wall bathtub bumper covers the front wall inner corner and at least a portion of the front wall interior surface.

[0021] In another aspect of the invention, a bathtub bumper comprises a front wall bathtub bumper that is sized and shaped to be positionable on a front wall rim portion of a bathtub, wherein the front wall bathtub bumper covers at least one inch or at least two inches of the front wall exterior surface for at least about 50%, 60%, 70%, 80%, or 90% of the length of the front wall exterior surface.

[0022] In another aspect of the invention, a bathtub bumper comprises a front wall bathtub bumper that is sized and shaped to be positionable on a front wall rim portion of a bathtub, wherein the front wall bathtub bumper is molded or otherwise formed into a generally preset shape.

[0023] In another aspect of the invention, a bathtub bumper comprises a front wall bathtub bumper that is sized and shaped to be positionable on a front wall rim portion of a bathtub, wherein the front wall bathtub bumper is molded or otherwise formed into a generally preset shape, wherein the preset shape generally corresponds to the bathtub front wall rim that the front wall bathtub bumper is to be positioned on.

[0024] In another aspect of the invention, a bathtub bumper comprises a front wall bathtub bumper that is sized and shaped to be positionable on a front wall rim portion of a bathtub, wherein the front wall bathtub bumper is molded or otherwise formed into a generally preset shape, wherein the preset shape includes a U-shaped channel.

[0025] In another aspect of the invention, a bathtub is molded or otherwise formed into a generally preset shape, wherein the preset shape includes an L-shaped channel.

[0026] In another aspect of the invention, a method of using a bathtub bumper comprises positioning a bathtub bumper in accordance with any of the aspects discussed above or disclosed herein on a bathtub.

[0027] In another aspect of the invention, a bathtub bumper is provided for a bathtub, the bathtub comprising a front wall, the front wall having a front wall interior surface, a front wall exterior surface, and a front wall rim portion connecting the front wall interior surface and the front wall exterior surface, the bathtub bumper comprising: a body comprising cushioning material having a generally preset shape, the body having a top portion, an interior sidewall portion extending downwardly from the top portion, and an exterior sidewall portion extending downwardly from the top portion, wherein the top portion, the interior sidewall portion, and the exterior sidewall portion form a U-shaped channel adapted to be positioned on the front wall of the bathtub so that the top portion covers at least a portion of the front wall rim portion, the interior sidewall portion covers at least a portion of the front wall interior surface, and the

exterior sidewall portion covers at least a portion of the front wall exterior surface, wherein the bathtub bumper is removeably positionable on the front wall of the bathtub in a manner that decreases the risk or severity of injuries in and around a bathtub.

[0028] In another aspect of the invention, a bathtub bumper is provided for a bathtub, the bathtub comprising a front wall, the front wall having a front wall interior surface, a front wall exterior surface, and a front wall rim portion connecting the front wall interior surface and the front wall exterior surface, the bathtub bumper comprising: a body comprising cushioning material having a generally preset shape, the body having a top portion, an interior sidewall portion extending downwardly from the top portion, and an exterior sidewall portion extending downwardly from the top portion, wherein the top portion, the interior sidewall portion, and the exterior sidewall portion form a U-shaped channel adapted to be positioned on the front wall of the bathtub so that the top portion covers at least a portion of the front wall rim portion, the interior sidewall portion covers at least a portion of the front wall interior surface, and the exterior sidewall portion covers at least a portion of the front wall exterior surface, wherein the bathtub bumper is removeably positionable on the front wall of the bathtub in a manner that decreases the risk or severity of injuries in and around a bathtub, and wherein the interior sidewall portion extends downwardly from an interior surface of the top portion at least 0.5 inches so that when positioned on the front wall, at least 0.5 inches of the front wall interior surface is covered, and wherein the exterior sidewall portion extends downwardly from the interior surface of the top portion at least 0.5 inches so that when positioned on the front wall, at least 0.5 inches of the front wall exterior surface is covered.

[0029] In another aspect of the invention, a bathtub bumper is provided for a bathtub, the bathtub comprising a front wall, the front wall having a front wall interior surface, a front wall exterior surface, and a front wall rim portion connecting the front wall interior surface and the front wall exterior surface, the bathtub bumper comprising: a body comprising cushioning material having a generally preset shape, the body having a top portion, an interior sidewall portion extending downwardly from the top portion, and an exterior sidewall portion extending downwardly from the top portion, wherein the top portion, the interior sidewall portion, and the exterior sidewall portion form a U-shaped channel adapted to be positioned on the front wall of the bathtub so that the top portion covers at least a portion of the front wall rim portion, the interior sidewall portion covers at least a portion of the front wall interior surface, and the exterior sidewall portion covers at least a portion of the front wall exterior surface, wherein the bathtub bumper is removeably positionable on the front wall of the bathtub in a manner that decreases the risk or severity of injuries in and around a bathtub, and wherein the interior sidewall portion extends downwardly from an interior surface of the top portion from about 0.5 inches to about 8 inches so that when positioned on the front wall, from about 0.5 inches to about 8 inches of the front wall interior surface is covered, and wherein the exterior sidewall portion extends downwardly from the interior surface of the top portion at least 0.5 inches so that when positioned on the front wall, at least 0.5 inches of the front wall exterior surface is covered.

[0030] In another aspect of the invention, a bathtub bumper system is provided for a bathtub, the bathtub comprising a front wall and a second wall, the front wall having a front wall interior surface, a front wall exterior surface, and a front wall rim portion connecting the front wall interior surface and the front wall exterior surface, and the second wall comprising a second wall interior surface and a second wall rim portion, the bathtub bumper system comprising: a first body comprising cushioning material, the body having a top portion, an interior sidewall portion extending downwardly from the top portion, and an exterior sidewall portion extending downwardly from the top portion, wherein the top portion, the interior sidewall portion, and the exterior sidewall portion form a U-shaped channel adapted to be positioned on the front wall of the bathtub so that the top portion covers at least a portion of the front wall rim portion, the interior sidewall portion covers at least a portion of the front wall interior surface, and the exterior sidewall portion covers at least a portion of the front wall exterior surface, and a second body comprising cushioning material, the second body being shaped differently than the first body, and the second body having a top portion and an interior sidewall portion extending downwardly from the top portion, wherein the top portion and the interior sidewall portion form a channel adapted to be positioned on the second wall of the bathtub so that the top portion covers at least a portion of the second wall rim portion and the interior sidewall portion covers at least a portion of the second wall interior surface, wherein the first body is removeably positionable on the front wall of the bathtub and the second body is removeably positionable on the second wall of the bathtub in a manner that decreases the risk or severity of injuries in and around a bathtub.

[0031] In another aspect of the invention, a method of preventing bathtub injuries, comprises positioning a bathtub bumper on a front wall of a bathtub, the front wall having a front wall interior surface, a front wall exterior surface, and a front wall rim portion connecting the front wall interior surface and the front wall exterior surface, wherein the bathtub bumper comprises a body comprising cushioning material having a preset shape, the body having a top portion, an interior sidewall portion extending downwardly from the top portion, and an exterior sidewall portion extending downwardly from the top portion, wherein the top portion, the interior sidewall portion, and the exterior sidewall portion form a U-shaped channel adapted to be positioned on the front wall of the bathtub so that the top portion covers at least a portion of the front wall rim portion, the interior sidewall portion covers at least a portion of the front wall interior surface, and the exterior sidewall portion covers at least a portion of the front wall exterior surface.

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] These features, aspects, and advantages of the present invention will become better understood with regard to the following description, appended claims, and accompanying drawings which illustrate exemplary features of the invention. However, it is to be understood that each of the features can be used in the invention in general, not merely in the context of the particular drawings, and the invention includes any combination of these features, where:

[0033] FIG. 1A is a schematic perspective view of a version of a bathtub bumper of the invention positioned on a front rim of a bathtub;

[0034] FIG. 1B is a schematic sectional view of the bathtub bumper of FIG. 1A in the 1B-1B direction;

[0035] FIG. 2A is a schematic perspective view of a version of a bathtub bumper system of the invention positioned on a plurality of rims of a bathtub;

[0036] FIG. 2B is a schematic sectional view of the bathtub bumper system of FIG. 2A in the 2B-2B direction;

[0037] FIG. 3A is a schematic perspective view of another version of a bathtub bumper system of the invention positioned on three rims of a bathtub;

[0038] FIG. 3B is a schematic sectional view of the bathtub bumper system of FIG. 3A in the 3B-3B direction;

[0039] FIG. 4A is a schematic perspective view of another version of a bathtub bumper system of the invention positionable on a rim of a bathtub;

[0040] FIG. 4B is a schematic top view of the bathtub bumper system of FIG. 4A;

[0041] FIG. 5A is a schematic front view of the bathtub bumper system of FIG. 4A;

[0042] FIG. 5B is a schematic rear view of the bathtub bumper system of FIG. 4A;

[0043] FIG. 5C is a schematic left side view of the bathtub bumper system of FIG. 4A;

[0044] FIG. 6A is a schematic bottom view of another version of a bathtub bumper system of the invention positionable on a rim of a bathtub;

[0045] FIG. 6B is a schematic closeup view of a recess in the bathtub bumper system of FIG. 6A;

[0046] FIG. 6C is a schematic closeup view of a recess with a suction cup in the bathtub bumper system of FIG. 6A;

[0047] FIG. 6D is a schematic partially sectional view of a portion of a version of a bathtub bumper system with a replaceable suction cup in a recess;

[0048] FIG. 7A is a schematic perspective view of the bathtub bumper system of FIG. 4A positioned on a bathtub; and

[0049] FIG. 7B is a schematic sectional view of the bathtub bumper system of FIG. 7A in the 7B-7B direction.

DESCRIPTION

[0050] The present invention relates to a bathtub bumper. In particular, the invention relates to a bathtub bumper that is positionable on a bathtub rim. Although the bathtub bumper is illustrated and described in the context of being useful alone or in concert with other bathtub bumpers for providing protection against injury in a bathroom, the present invention can be useful in other instances. Accordingly, the present invention is not intended to be limited to the examples and embodiments described herein.

[0051] FIGS. 1A and 1B show a version of a bathtub bumper 100 of the invention. The bathtub bumper is positionable on a bathtub 105 in a manner that it protectively covers one or more potentially dangerous surfaces on the bathtub 105. The bathtub 105 is made up of a tub cavity 110 that is designed and adapted to contain bath water for a user taking a bath and/or to be stood in or sat in for a user taking a shower. The tub cavity 110 is defined by a bottom surface 115 that typically includes a drain, sidewalls 120, including a front wall 121, a rear wall 122, a left wall 123, and a right wall 124. The tub cavity also has an open top portion 125 through which a user accesses the tub cavity 110. The front wall 121 and the rear wall 122 typically extending in a longitudinal direction and with the left wall 123 and right wall 124 typically being shorter and extending in a direction

orthogonal to the longitudinal direction. A faucet or the like is typically provided on the left wall **123** or the right wall **124** but may also or alternatively be provided elsewhere. The sidewalls **120** include interior surfaces **130** including a front wall interior surface **131**, a rear wall interior surface **132**, a left wall interior surface **133**, and a right wall interior surface **134**. At least a portion of the interior surfaces **130** helps to form the tub cavity **110**. The interior surfaces **130** extends upwardly to a rim portion **135** that is associated with the open top portion **125** of the tub cavity **110**. One or more of the sidewalls **120**, most typically the front wall **121**, also includes one or more exterior surfaces **140**, such as the front wall exterior surface **141**, that is on the opposite side of the front wall **121** from the interior front wall interior surface **131** and the tub cavity **110**. The front wall interior surface **131** and the front wall exterior surface **141** rise to meet at the front wall rim portion **145**. Typically, the front wall rim portion **145** will include an inner corner or edge **150** and an outer corner or edge **155**. The front wall inner corner **150** and the front wall outer corner **155** of the front wall rim portion **145** are often connected by a front wall ledge portion **160** which is typically from about 2 inches to about 8 inches wide. The front wall ledge portion **160** may include a flat portion and/or an at least partially rounded portion. In the case where the inner corner or edge **150** and/or the outer corner or edge **155** is rounded, the front wall rim portion **145** will be considered to transition to the front wall interior surface **131** or the front wall exterior surface **141** at the point where a tangent line extending orthogonally across the front wall **121** exceeds 45 degrees or becomes more vertical than horizontal. To gain access to the tub cavity **110**, a user steps over, crawls over, or is lifted over the front wall rim portion **145**.

[0052] In the version of FIGS. 1A and 1B, the bathtub bumper **100** is a front wall bathtub bumper **165** that is sized and shaped to be positionable on the front rim portion **145** of the front wall **121** of a bathtub **105**. The front wall bathtub bumper **165** includes a body **170** made of cushioning material. The body **170** of the front wall bathtub bumper **165** includes a top portion **171**, an interior sidewall portion **172** adapted to be placed over at least a portion of the front wall interior surface **131**, an exterior sidewall portion **173** opposite the interior sidewall portion **172** and adapted to be placed over at least a portion of the front wall exterior surface **141**, and a bottom portion **174**. A bathtub receiving channel **175** extends into the bottom portion **171** of the body **170**. In the version of FIGS. 1A and 1B, the bathtub receiving channel **175** is a U-shaped channel **180** that extends into the bottom portion **174** to define the top portion **171**, the interior sidewall portion **172**, and the exterior sidewall portion **173**. The front wall bathtub bumper **165** is positionable on the front wall **121** by installing the U-shaped channel **180** over the front wall rim portion **145** so that the top portion **171** covers the front wall ledge portion **160**, the interior sidewall portion **172** covers at least a portion of the front wall interior surface **131**, and the exterior sidewall portion **173** covers at least a portion of the front wall exterior surface **141**. By U-shaped channel it is meant any channel that includes two generally opposing sides connected by a base portion with the generally opposing sides extending away from the base portion in generally the same direction. In the version of FIGS. 1A and 1B, the U-shaped channel **180** is an inverted U-shaped channel with the top portion **171** of the body **170** serving as the base and the interior sidewall

portion **172** and exterior sidewall portion **173** serving as the generally opposing sides extending from the base. In the particular version shown, the interior sidewall portion **172** and/or the exterior sidewall portion **173** extend at about a 90 degree angle relative to the horizontal from the top portion **171**, but the angle can be varied to any desired angle to suit the shape of the bathtub **105** the bathtub bumper **100** is being positioned on. For example, in one version, the angles can range from about 30 degrees to about 120 degrees or from about 75 degrees to about 100 degrees. Similarly, in the version shown, the top portion **171** is shown to have a flat portion to match the flat shape of the front wall ledge portion **160**. However, the interior and/or exterior of the top portion **171** can be differently shaped to match the shape of the front wall ledge portion **160** or to provide a differently shaped surface from the front wall ledge portion **160**.

[0053] The bathtub bumper **100**, such as the front wall bathtub bumper **165** of FIGS. 1A and 1B, is designed to cover at least a portion of a hard surface of a bathtub **105** to prevent or reduce the severity of an injury that results from a slip, fall, or drop of an individual in or around the bathtub **105**. In the version shown, the bathtub bumper **100** comprises a cushioning material that is adapted to provide a cushioning effect when an object, such as a body part of an individual, contacts a hard surface of the bathtub **105**. For example, in one version, the bathtub bumper **100** comprises a material that is softer and/or more compressible than the material of the bathtub **105** it is covering. When an object contacts the bathtub bumper **100** when it is positioned to cover a portion of the bathtub **100**, the bathtub bumper **100** will compress to absorb some of the energy of the object's movement and/or to make the deceleration of the object less abrupt and thereby lessen the severity of the impact when compared to a similar impact being made in the absence of the bathtub bumper **100**. For example, the bathtub bumper **100** can comprise one or more of foam, rubber, molded soft plastic, cork, and the like, and in one particular version comprises thermoplastic elastomer foam and/or ethylene-vinyl acetate foam. The bathtub bumper can have a thickness selected to provide a desired cushioning effect. In one version the thickness, T, as shown for example in FIG. 1B, is from about 1 inch to about 5 inches, or from about 1.5 inches to about 4 inches, or from about 2 inches to about 3.5 inches, or about 2.25 inches. The thickness can vary depending on the material being used with a softer material having a larger thickness.

[0054] The size and/or length of the bathtub bumper **100** can also be selected to provide a desired protection to a hard bathtub surface and/or to provide additional benefits. The version of the front wall bathtub bumper **165** of FIGS. 1A and 1B is sized to protect against injury in multiple ways. For example, the front wall bathtub bumper **165** has a sufficient longitudinal length from a left end **185** to a right end **190** to cover a significant portion of the front wall rim portion **145**. In one version, the length, L, from the left end **185** to the right end **190** is from about 40 inches to about 70 inches, or from about 45 inches to about 60 inches, or about 50 to 55 inches, or about 55 inches. In one version, the length of the front wall bathtub bumper **165** from the left end **185** to the right end **190** is sufficient to cover at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90%, or at least about 95%, or about 100% of the length of the front wall rim portion **145**. By covering a significant portion of the front wall rim

portion **145**, the front wall bathtub bumper **165** provides improved coverage of protection over an object such as a towel. A slip, fall, or drop that occurs in the area of the front wall **121** of the bathtub **105** can result in a part of a body contacting front wall rim portion **145** at any point within a several-foot radius of where the slip, fall, or drop started. For example, a person stepping into or out of the bathtub **105** near the right wall **124** can slip in a manner that results in their head hitting the front wall rim portion **145** at a position near the left wall **123**. Accordingly, by covering a significant portion of the front wall rim portion **145** there is an increased likelihood that the location of contact will be covered and protected.

[0055] The shape and/or design of the bathtub bumper **100** can also be selected to provide a desired protection to a hard bathtub surface and/or to provide additional benefits. The version of the front wall bathtub bumper **165** of FIGS. **1A** and **1B** is designed to protect against injury in multiple ways. As can be seen in FIG. **1B**, the front wall bathtub bumper **165** is shaped so that both the front wall inner corner **150** and the front wall outer corner **155** are covered by the bathtub bumper **100**. The hard corner edges of the front wall inner corner **150** and the front wall outer corner **155** when not covered by the bathtub bumper **100** can be particularly dangerous and injurious. By adequately covering both the front wall inner corner **150** and the front wall outer corner **155**, protection is provided from slips, fall, or drops that occur while inside the tub cavity **110**, while outside the tub cavity **110**, and while step into or out of the tub cavity **110**. In addition, in one version, the front wall bathtub bumper **165** covers at least a portion of the front wall interior surface **131** and/or the front wall exterior surface **141** assures that the entireties of the front wall inner corner **150** and/or the front wall outer corner **155** are covered and also provides cushioning protection to the top of the interior and/or exterior surfaces which themselves can be the source of injuries. In one particular version, the front wall bathtub bumper **165** covers at least a portion of both the front wall interior surface **131** and the front wall exterior surface **141**. This version not only provides the advantages described above but provides the additional advantage of helping to secure the front wall bathtub bumper **165** onto the front wall **121**. By contacting both the front wall interior surface **131** and the front wall exterior surface **141**, the front wall rim portion **145** can be sandwiched between the interior sidewall portion **172** and the exterior sidewall portion **173** of the body **170** of the front wall bathtub bumper **165**. This arrangement helps to prevent the front wall bathtub bumper **165** from being accidentally, inadvertently, or unintentionally pulled or kicked off. Accordingly, in one version of the front wall bathtub bumper **165**, the body **170** has an interior sidewall portion **172** and/or an exterior sidewall portion **174** that is designed to cover the front wall interior surface **131** and/or the front wall exterior surface **141** by of height, *H*, of at least about 0.5 inches, or at least about 1.0 inches, or at least about 2 inches. In addition, in one version, the height, *H*, of the interior sidewall portion **172** can be limited so as to avoid extending so far down the front wall interior surface **131** that the interior sidewall portion **172** becomes too wet and/or begins obstructing the space within the tub cavity **110**. Accordingly, in this version, the front wall interior sidewall portion has a

height, *H*, of from about 0.5 inches to about 8 inches, or from about 1 inch to about 5 inches, or from about 2 inches to about 4.5 inches.

[0056] In one version, the bathtub bumper **100**, such as the front wall bathtub bumper **165** of FIGS. **1A** and **1B**, has a body **170** that is molded or otherwise formed into a generally preset shape, such as the shape shown in FIGS. **1A** and **1B**. The generally preset shape is a shape that generally corresponds to the bathtub **105** at the position of attachment, which in the case of the front wall bathtub bumper **165** would be the front wall **121** of the bathtub **105**. By generally preset shape it is meant that the bathtub bumper **165** substantially takes on and maintains its desired shape when not acted on by outside forces. The desired shape can be, for example, its in-use shape or an installation shape. One particular example, of an installation shape might be where the U-shaped channel **180** of the front wall bathtub bumper **165** includes an inwardly angled interior sidewall portion **172** and/or exterior sidewall portion **173**, and wherein the sidewalls can be spread apart during installation on the front wall **121** so that they squeeze against front wall **121** to help secure the front wall bathtub bumper **165** on the front wall **141**. In one version, the generally preset shape can be molded or deformed when acted on by an outside force. For example, the bathtub bumper **100** can be molded or pressed into a fitting a particular contour of the bathtub **105** and/or can be rolled or pressed into a different shape for storage or shipping and then return to its preset shape with the force is removed. The left end **185** and/or the right end **190** of the bathtub bumper **100** can optionally also have an end contour **195**, such as a preset or moldable curvature that allows the left end **185** and/or the right end **190** to curve along the bathtub rim or to engage or nearly engage another bathtub bumper, as will be discussed.

[0057] By providing the front wall bathtub bumper **165** with a body **170** made of cushioning material in a generally shape, the bathtub bumper **100** of the invention has several advantages over previous bathtub safety devices. For example, the preset shape provides better protection at the inner corner edge **150** and/or the outer corner edge **155** of the bathtub **105** than a device that lacks a preset shape. For example, devices that are flexible mats or pads are sufficiently flexible to be bent around the corners, and this flexibility makes the device vulnerable to injury in a corner region. By having a generally preset shape, the bathtub bumper **100** of the present invention does not have to be bendable and can more focus can be applied to making the corner region more protective when there is contact in that region. In addition, as discussed above, the preset shape can be used to attach the bathtub bumper **100** to the bathtub **105** by having the preset shape correspond to the size and shape of a portion of the bathtub **105**, such as the front wall **121**, as shown in FIG. **1B**. Optionally, the preset shape can be designed to compress, squeeze, or clamp onto the front wall **121** so that the front wall **121** is compressingly sandwiched within the U-shaped channel **180**. The type of attachment has several advantages. For example, with pads that are draped over a front wall **121**, there are spaces or gaps between the pad and the bathtub sidewalls **120**, and these spaces can be the source of mold growth or other uncleanliness. Pads that have sidewall suction cups but with a body that lacks a preset shape are particularly susceptible to mold growing spaces. With the preset shape of the bathtub bumper **100** of the invention, these gaps can be reduced and/or the

number of suction cups necessary to secure the bathtub bumper 100 can be reduced or eliminated and/or the suction cups can be located in positions less likely to be exposed to water, such as on the top portion 171 or on the exterior sidewall portion 173. Furthermore, by having a preset shape, the bathtub bumper 100 can be sized and shaped to correspond to a particular sized and shaped bathtub 105, and in this case the bathtub bumper 100 can be more easily and/or properly positioned on the bathtub 105.

[0058] FIGS. 2A and 2B show a bathtub bumper system 200 according to another version of the invention. The bathtub bumper system 200 comprises a plurality of bathtub bumpers 100, such as a front wall bathtub bumper 165, as described above, and one or more additional bathtub bumpers. The bathtub bumper system 200 thus provides a modular system whereby a plurality portions of the rim portion 135 can be protected, such as the front wall rim portion 145 and one or more of the rear wall rim portion 205, the left wall rim portion 210, and the right wall rim portion 215. In the particular version of FIGS. 2A and 2B, the bathtub bumper system 200 comprises four bathtub bumpers 100, including a front wall bathtub bumper 165, a rear wall bathtub bumper 220, a left wall bathtub bumper 225, and a right wall bathtub bumper 230.

[0059] In one version, the one or more additional bathtub bumpers is a bathtub bumper shaped similarly to the front wall bathtub bumper 165. In another version, such as the one shown in FIGS. 2A and 2B, the one or more additional bathtub bumpers is a single corner bathtub bumper 235. Sometimes, a bathtub 105 is designed where one or more of the sidewalls 120 has no exterior corner or the bathtub 105 is positioned in a location where an exterior corner is of little threat. For such sidewalls 120 it may be desirable to provide a bathtub bumper 100 as a single corner bathtub bumper 235. In the particular version of FIGS. 2A and 2B, each of the rear wall bathtub bumper 220, the left wall bathtub bumper 225, and the right wall bathtub bumper 230 are single corner bathtub bumpers 235. However, in another version, only one or two of these are single corner bathtub bumpers 235. As can be seen in FIG. 2B, with a single corner bathtub bumper 235, the bathtub receiving channel 175 is an L-shaped channel 240. By L-shaped channel it is meant the same as the U-shaped channel discussed above but with the exterior sidewall portion. The single corner bathtub bumper 235 thus covers and provides protection on the rear wall internal corner 245 along at least a portion of a rear wall top surface 250 and the rear wall interior surface 132. Similarly, the left wall bathtub bumper 225 and the right wall bathtub bumper 230 cover the respective internal corners, tops, and interior surfaces of the left wall 123 and the right wall 124. Since the rear wall 122, the left wall 123, and the right wall 124 are typically not stepped over, there is less chance of contact and dislodgement of bathtub bumpers 100 positioned thereon. The materials, thicknesses, and coverages of the single corner bathtub bumper 235 can be the same or similar to the ones discussed above for the front wall bathtub bumper 165. The lengths of the rear wall bathtub bumper 220 can be the same as for the front wall bathtub bumper 165, and the lengths of the left wall bathtub bumper 225 and the right wall bathtub bumper 230 can be proportionately the same as for the front wall bathtub bumper 165 relative the lengths of the left wall 123 and right wall 124 when compared to the front wall 121. The end contours 195 of

each of the bathtub bumpers 100 can be designed to correspond to one another so that they fit together or provide a generally continuous profile.

[0060] The modularity of the bathtub bumper system 200, such as the one shown in FIGS. 2A and 2B, offers several advantages. For example, the bathtub bumper system 200 allows for coverage of all dangerous internal corners in the bathtub 105. In addition, the bathtub bumper system 200 is easy to ship, store, and install. The modularity also allows the bathtub bumper system 200 to fit or conform to multiple shaped bathtubs 105. Each of the sidewalls 120 can be equipped with an appropriately shaped bathtub bumper 100 for its shape and/or threat of danger. The individual bathtub bumpers 100 can be separately removed to be cleaned or replaced. Furthermore, a space or gap 255 can be provided between adjacent bathtub bumpers 100 that can allow air to pass and circulate between the bathtub bumpers 100 to help with drying the bathtub bumpers 100 when they are installed on the bathtub 105.

[0061] As with the front wall bathtub bumper 165, the sizes and shapes of the rear wall bathtub bumper 220, the left wall bathtub bumper 225, and/or the right wall bathtub bumper 230 can be selected to suit the desired purpose. For example, the thickness of the rear wall bathtub bumper 220, the left wall bathtub bumper 225, and/or the right wall bathtub bumper 230 can be the same as the thickness above for the front wall bathtub bumper 165. Each of the rear wall bathtub bumper 220, the left wall bathtub bumper 225, and/or the right wall bathtub bumper 230, and the front wall bathtub bumper 165 can be the same or one or more can be different. The lengths of the rear wall bathtub bumper 220 can be the same or similar to the lengths discussed above for the front wall bathtub bumper 165. The lengths of the left wall bathtub bumper 225 and/or the right wall bathtub bumper 230 proportionately the same as for the front wall bathtub bumper 165 discussed above but adjusted for the relative different in length of the left wall rim portion 205 and/or right wall rim portion 210 when compared with the front wall rim portion 145 of a particular bathtub 105.

[0062] Also, as with the front wall bathtub bumper version discussed above, the rear wall bathtub bumper 220, the left wall bathtub bumper 225, and/or the right wall bathtub bumper 230 can be molded or otherwise formed into a generally preset shape, such as the shape shown in FIGS. 2A and 2B. The generally preset shape is a shape that generally corresponds to the bathtub 105 at the position of attachment. In one version, the preset shape includes an L-shaped channel. As with the front wall bathtub bumper version discussed above, the preset shape can be molded or deformed when acted on by an outside force. For example, the bathtub bumper 100 can be molded or pressed into a fitting a particular contour of the bathtub 105 and/or can be rolled or pressed into a different shape for storage or shipping and then return to its preset shape with the force is removed.

[0063] FIGS. 3A and 3B show another version of the bathtub bumper system 200 that is similar to the version of FIGS. 2A and 2B. In the version of FIGS. 3A and 3B, one or more of the sidewalls 120, such as the right wall 124 has no internal corner. Instead, the right wall 124 is a smooth continuation of the wall of the bathroom or shower. With this type of bathtub 105, there is no need for a bathtub bumper 100 along the cornerless sidewall. Accordingly, in the version of FIGS. 3A and 3B, there are three bathtub bumpers

100 provided. In the particular version shown, the bathtub bumper system 200 comprises a front wall bathtub bumper 165 and two single corner bathtub bumpers 235.

[0064] FIGS. 4A, 4B, 5A, 5B, and 5C show another version of the bathtub bumper system 200 that is similar to the version of FIGS. 2A and 2B. In this version, one or more of the bathtub bumpers 100, such as the front wall bathtub bumper 165 includes one or more bathing related features 400. For example, one bathing related feature 400 is a drainage system 405 that includes one or more drainage channels 410 in the top portion 171 of the body 170 of the bathtub bumper 100. The drainage system 405 is designed and shaped to help direct water that contacts the top portion 170 back into the tub cavity 110. Another bathing related feature 400 is one or more storage recesses 415 provided on the body 170 of the bathtub bumper 100. The one or more storage recessed can be used to conveniently store or hold a shampoo bottle, toy, or any other item that may be of use while bathing. Optionally, the drainage channels 410 can be sloped so that the bottom of the draining channel 410 at a forward end is higher than the bottom of the drainage channel at a rearward end so that water flow towards the tub cavity 110 when the bathtub bumper 100 is positioned on a bathtub 105.

[0065] FIGS. 6A, 6B, and 6C show another version of the bathtub bumper system 200 that is similar to the version of FIGS. 2A and 2B. In the version of FIGS. 6A, 6B, and 6C, one or more the bathtub bumpers 100 and/or each of the bathtub bumper 100 of a bathtub bumper system 200 include an additional attachment mechanism 600. The additional attachment mechanism 600 can be used in addition to or in place of the attachment of the bathtub bumper 100 by virtue of its preset shape, as discussed above. For example, in the version shown, the additional attachment mechanism 600 comprises one or more suction mechanisms 605 as can be seen on the underside surface 610 of the bathtub bumpers 100. The attachment mechanism allows the bathtub bumper 100 to be more securely held in position on the bathtub 105 or to hold the bathtub bumper 100 in place when a preset shape of the bathtub bumper 100 is not a perfect fit on the bathtub 105. FIG. 6B shows a close-up view of a version of the underside surface 610 in the region of a suction mechanism 605. In this version, the suction mechanism 605 comprises a suction cup recess 615 that is sized and shaped to receive a suction cup 620 within the suction cup recess 615 provided on the underside surface 610. FIG. 6C shows a suction cup 620 in the recess 615. As can be seen in FIG. 6C, in this version, the suction mechanism 605 is designed so that the suction cup 620 is substantially flush with the underside surface 610 of the bathtub bumper 100. By being substantially flush, the suction mechanism 605 does not project away from the underside surface 610 and thus reduces the number or size of mold growing gaps and spaces between the underside surface 610 and the bathtub sidewalls 120.

[0066] Optionally, as shown in the version of FIG. 6D, the suction cup 620 can be a replaceable suction cup 625. In this version, the replaceable suction cup 625 has a stem portion 630 that is receiveable within a receiving member 635 in the suction cup recess 615. The stem portion 630 clicks in to position in the receiving member 635 such as by a having a stem protrusion 640 or the like lockingly engage a receiving member protrusion 645 or recess or the like. The version of FIG. 6D allows for the periodic replacement of suction cups

625 when one becomes worn or moldy. Alternatively or additionally, the additional attachment mechanism 600 can comprise an adhesive, an adhesive strip, a tacky substance, and/or the like.

[0067] FIGS. 7A and 7B show a version of the bathtub bumper system 200 that includes the features of the version of FIGS. 4A, 4B, 5A, 5B, and 5C and the features of FIGS. 6A, 6B, and 6C positioned on a bathtub 105.

[0068] Although the present invention has been described in considerable detail with regard to certain preferred versions thereof, other versions are possible, and alterations, permutations and equivalents of the versions shown will become apparent to those skilled in the art upon a reading of the specification and study of the drawings. For example, the cooperating components may be reversed or provided in additional or fewer number, and all directional limitations, such as up and down and the like, can be switched, reversed, or changed as long as doing so is not prohibited by the language herein with regard to a particular version of the invention. Like numerals represent like parts from figure to figure. When the same reference number has been used in multiple figures, the discussion associated with that reference number in one figure is intended to be applicable to the additional figure(s) in which it is used, so long as doing so is not prohibited by explicit language with reference to one of the figures. Also, the various features of the versions herein can be combined in various ways to provide additional versions of the present invention. Furthermore, certain terminology has been used for the purposes of descriptive clarity, and not to limit the present invention. Throughout this specification and any claims appended hereto, unless the context makes it clear otherwise, the term “comprise” and its variations such as “comprises” and “comprising” should be understood to imply the inclusion of a stated element, limitation, or step but not the exclusion of any other elements, limitations, or steps. Throughout this specification and any claims appended hereto, unless the context makes it clear otherwise, the term “consisting of” and “consisting essentially of” should be understood to imply the inclusion of a stated element, limitation, or step and the exclusion of any other elements, limitations, or steps or the exclusion of any other essential elements, limitations, or steps, respectively. Throughout the specification, any discussion of a combination of elements, limitations, or steps should be understood to include (i) each element, limitation, or step of the combination alone, (ii) each element, limitation, or step of the combination with any one or more other element, limitation, or step of the combination, (iii) an inclusion of additional elements, limitations, or steps (i.e. the combination may comprise one or more additional elements, limitations, or steps), and/or (iv) an exclusion of additional elements, limitations, or steps or an exclusion of essential additional elements, limitations, or steps (i.e. the combination may consist of or consist essentially of the disclosed combination or parts of the combination). All numerical values, unless otherwise made clear in the disclosure or prosecution, include either the exact value or approximations in the vicinity of the stated numerical values, such as for example about +/- ten percent or as would be recognized by a person or ordinary skill in the art in the disclosed context. The same is true for the use of the terms such as about, substantially, and the like. Also, for any numerical ranges given, unless otherwise made clear in the disclosure, during prosecution, or by being explicitly set forth in a

claim, the ranges include either the exact range or approximations in the vicinity of the values at one or both of the ends of the range. When multiple ranges are provided, the disclosed ranges are intended to include any combinations of ends of the ranges with one another and including zero and infinity as possible ends of the ranges. Therefore, any appended or later filed claims should not be limited to the description of the preferred versions contained herein and should include all such alterations, permutations, and equivalents as fall within the true spirit and scope of the present invention.

What is claimed is:

1. A bathtub bumper for a bathtub, the bathtub comprising a front wall, the front wall having a front wall interior surface, a front wall exterior surface, and a front wall rim portion connecting the front wall interior surface and the front wall exterior surface, the bathtub bumper comprising:

a body comprising cushioning material having a generally preset shape, the body having a top portion, an interior sidewall portion extending downwardly from the top portion, and an exterior sidewall portion extending downwardly from the top portion, wherein the top portion, the interior sidewall portion, and the exterior sidewall portion form a U-shaped channel adapted to be positioned on the front wall of the bathtub so that the top portion covers at least a portion of the front wall rim portion, the interior sidewall portion covers at least a portion of the front wall interior surface, and the exterior sidewall portion covers at least a portion of the front wall exterior surface,

wherein the bathtub bumper is removeably positionable on the front wall of the bathtub in a manner that decreases the risk or severity of injuries in and around a bathtub.

2. A bathtub bumper according to claim 1 wherein the front wall rim portion of the bathtub comprises a flat portion, and wherein the top portion of the body includes a flat portion adapted to match the flat portion of the front wall rim portion.

3. A bathtub bumper according to claim 1 wherein the interior sidewall portion extends from the top portion at a generally preset angle of from about 30 degrees to about 120 degrees and wherein the exterior sidewall portion extends from the top portion at a generally preset angle of from about 30 degrees to about 120 degrees.

4. A bathtub bumper according to claim 1 wherein the interior sidewall portion extends downwardly from an interior surface of the top portion at least 0.5 inches so that when positioned on the front wall, at least 0.5 inches of the front wall interior surface is covered, and wherein the exterior sidewall portion extends downwardly from the interior surface of the top portion at least 0.5 inches so that when positioned on the front wall, at least 0.5 inches of the front wall exterior surface is covered.

5. A bathtub bumper according to claim 1 wherein the interior sidewall portion extends downwardly from an interior surface of the top portion from about 0.5 inches to about 8 inches so that when positioned on the front wall, from about 0.5 inches to about 8 inches of the front wall interior surface is covered, and wherein the exterior sidewall portion extends downwardly from the interior surface of the top portion at least 0.5 inches so that when positioned on the front wall, at least 0.5 inches of the front wall exterior surface is covered.

6. A bathtub bumper according to claim 1 wherein the front wall extends in a longitudinal direction of the bathtub, and wherein the body is sized to cover at least 60 percent of the length of the front wall.

7. A bathtub bumper according to claim 1 wherein the front wall extends in a longitudinal direction of the bathtub, and wherein the body is sized to cover from about 70 percent to about 90 percent of the length of the front wall.

8. A bathtub bumper according to claim 1 wherein the cushioning material comprises a thermoplastic elastomeric foam that is from about 1 inch to about 5 inches thick.

9. A bathtub bumper according to claim 1 wherein the top portion comprises a top surface having one or more drainage channels therein.

10. A bathtub bumper according to claim 1 wherein one or more suction cups are provided in a recess in an underside surface of the body, wherein the one or more suction cups are substantially flush with the underside surface.

11. A bathtub bumper system for a bathtub, the bathtub comprising a front wall and a second wall, the front wall having a front wall interior surface, a front wall exterior surface, and a front wall rim portion connecting the front wall interior surface and the front wall exterior surface, and the second wall comprising a second wall interior surface and a second wall rim portion, the bathtub bumper system comprising:

a first body comprising cushioning material, the body having a top portion, an interior sidewall portion extending downwardly from the top portion, and an exterior sidewall portion extending downwardly from the top portion, wherein the top portion, the interior sidewall portion, and the exterior sidewall portion form a U-shaped channel adapted to be positioned on the front wall of the bathtub so that the top portion covers at least a portion of the front wall rim portion, the interior sidewall portion covers at least a portion of the front wall interior surface, and the exterior sidewall portion covers at least a portion of the front wall exterior surface, and

a second body comprising cushioning material, the second body being shaped differently than the first body, and the second body having a top portion and an interior sidewall portion extending downwardly from the top portion, wherein the top portion and the interior sidewall portion form a channel adapted to be positioned on the second wall of the bathtub so that the top portion covers at least a portion of the second wall rim portion and the interior sidewall portion covers at least a portion of the second wall interior surface,

wherein the first body is removeably positionable on the front wall of the bathtub and the second body is removeably positionable on the second wall of the bathtub in a manner that decreases the risk or severity of injuries in and around a bathtub.

12. A bathtub bumper system according to claim 11 wherein the first body and the second body have a generally preset shape.

13. A bathtub bumper system according to claim 11 wherein the bathtub comprises a third wall comprising a third wall interior surface and a third wall rim portion, and wherein the bathtub bumper system further comprises a third body comprising cushioning material, the third body being shaped different than the first body and the second body, and the third body having a top portion and an interior sidewall

portion extending downwardly from the top portion, wherein the top portion and the interior sidewall portion form a channel adapted to be positioned on the third wall of the bathtub so that the top portion covers at least a portion of the third wall rim portion and the interior sidewall portion covers at least a portion of the third wall interior surface.

14. A bathtub bumper system according to claim **11** wherein the channel of the second body is an L-shaped channel.

15. A bathtub bumper system according to claim **11** wherein the interior sidewall portion of the first body extends downwardly from an interior surface of the top portion of the first body at least 0.5 inches so that when positioned on the front wall, at least 0.5 inches of the front wall interior surface is covered, and wherein the exterior sidewall portion of the first body extends downwardly from the interior surface of the top portion of the first body at least 0.5 inches so that when positioned on the front wall, at least 0.5 inches of the front wall exterior surface is covered.

16. A bathtub bumper system according to claim **11** wherein the front wall extends in a longitudinal direction of the bathtub, wherein the second wall extends generally parallel to the front wall, wherein the first body is sized to cover at least 60 percent of the length of the front wall, and wherein the second body is sized to cover at least 60 percent of the length of the second wall.

17. A bathtub bumper system according to claim **11** wherein the front wall extends in a longitudinal direction of the bathtub, wherein the second wall extends generally orthogonal to the front wall, wherein the first body is sized to cover at least 60 percent of the length of the front wall,

and wherein the second body is sized to cover at least 60 percent of the length of the second wall.

18. A bathtub bumper system according to claim **11** wherein one or more suction cups are provided in a recess in an underside surface of the first body or the second body, wherein the one or more suction cups are substantially flush with the underside surface.

19. A method of preventing bathtub injuries, the method comprising:

positioning a bathtub bumper on a front wall of a bathtub, the front wall having a front wall interior surface, a front wall exterior surface, and a front wall rim portion connecting the front wall interior surface and the front wall exterior surface, wherein the bathtub bumper comprises a body comprising cushioning material having a preset shape, the body having a top portion, an interior sidewall portion extending downwardly from the top portion, and an exterior sidewall portion extending downwardly from the top portion, wherein the top portion, the interior sidewall portion, and the exterior sidewall portion form a U-shaped channel adapted to be positioned on the front wall of the bathtub so that the top portion covers at least a portion of the front wall rim portion, the interior sidewall portion covers at least a portion of the front wall interior surface, and the exterior sidewall portion covers at least a portion of the front wall exterior surface.

20. A method according to claim **19** further comprising positioning a bathtub bumper body on a second wall of the bathtub.

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